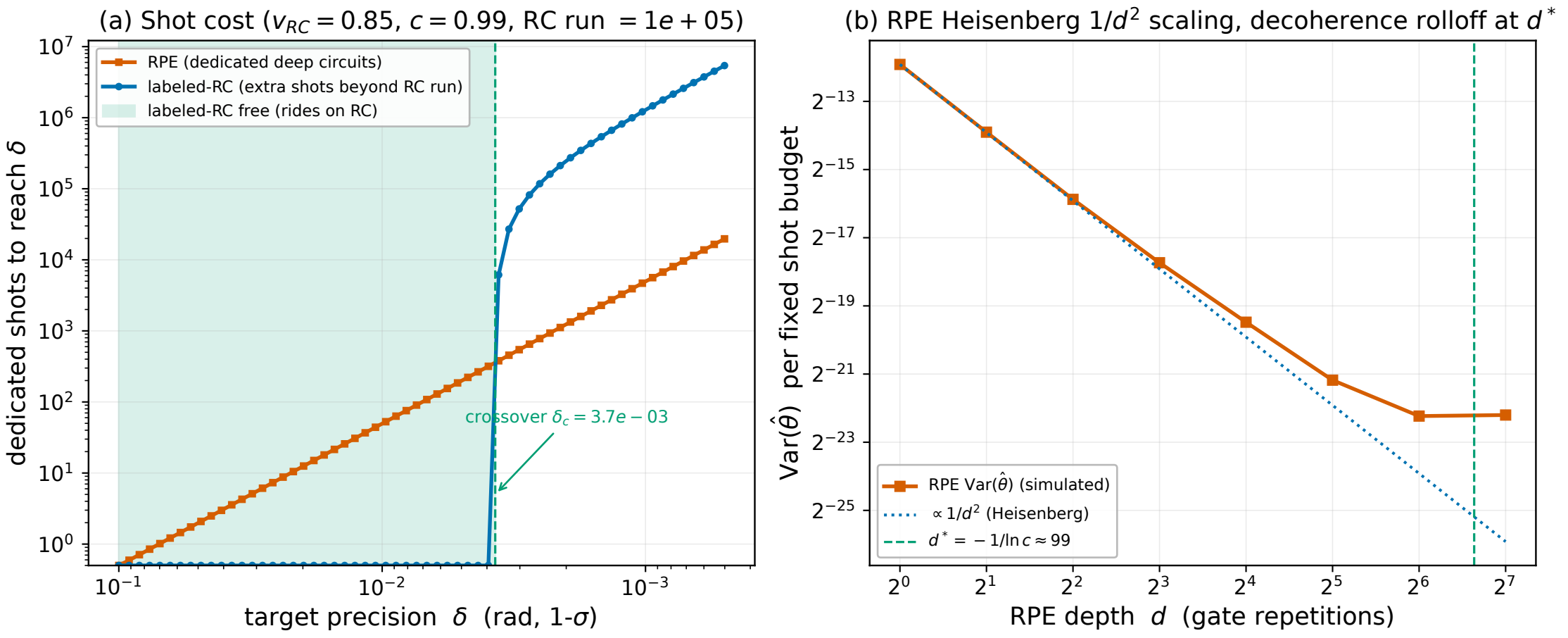
(a) Shot cost ($v_{RC} = 0.85$, $c = 0.99$, RC run $= 1e+05$)
RPE (dedicated deep circuits)
labeled-RC (extra shots beyond RC run)
labeled-RC free (rides on RC)
crossover $\delta_c = 3.7e-03$
dedicated shots to reach δ
target precision δ (rad, 1-σ)
(b) RPE Heisenberg $1/d^2$ scaling, decoherence rolloff at d^*
RPE Var($\hat{\theta}$) (simulated)
$\propto 1/d^2$ (Heisenberg)
$d^* = -1/\ln c \approx 99$
Var($\hat{\theta}$) per fixed shot budget
RPE depth d (gate repetitions)